

OpenShift Data Foundation (ODF) Deployment Guide

1. Overview

This document provides a production-grade procedure for deploying OpenShift Data Foundation (ODF) on a connected OpenShift Container Platform (OCP) cluster. It includes prerequisites, installation steps, validation, and troubleshooting guidance.

2. Architecture Overview

ODF provides the following storage services:

- Block Storage (RBD)
- File Storage (CephFS)
- Object Storage (NooBaa S3 compatible)

Core components:

- Ceph MON (Monitor)
- Ceph MGR (Manager)
- Ceph OSD (Storage Daemons)
- Ceph MDS (File system metadata)
- NooBaa (Object storage layer)

3. Prerequisites

3.1 OpenShift Cluster Requirements

- OCP version: 4.12 or later (recommended 4.14+)
- Minimum 3 worker nodes (for production HA)
- Cluster must be in healthy state

Validation:

```
oc get nodes
```

```
oc get clusterversion
```

3.2 Hardware Requirements

Each worker node must have:

- Minimum 16–32 GB RAM
- Minimum 4 CPUs
- Raw unformatted disks (SSD recommended)

Example disks:

- /dev/sdb
- /dev/sdc
- /dev/sdd

3.3 Network Requirements

- Full internal connectivity between nodes
- API and ingress reachable
- DNS resolution working for cluster domain

3.4 Operators Required

- OpenShift Data Foundation Operator
- Local Storage Operator

4. Pre-Checks

4.1 Node health check

```
oc get nodes -o wide
```

4.2 Cluster operators health

```
oc get co
```

5. Installing ODF Operator

Step 1: Open OperatorHub

Go to:

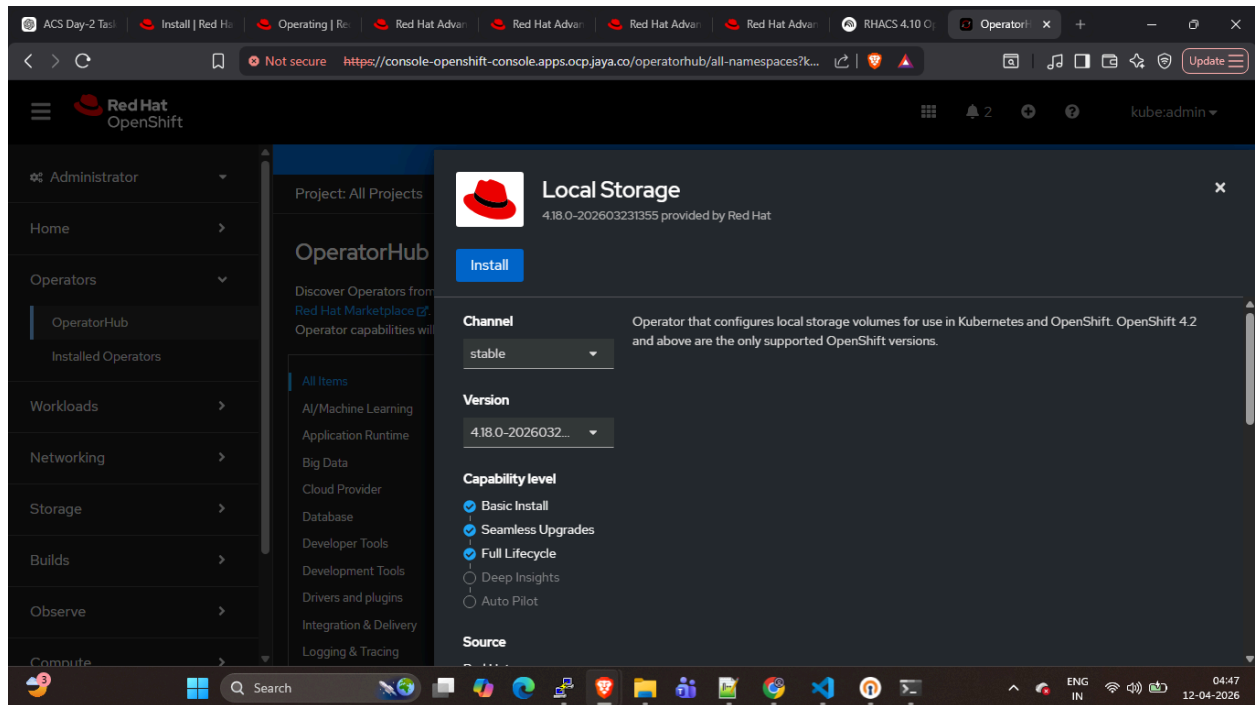
OpenShift Console → Operators → OperatorHub

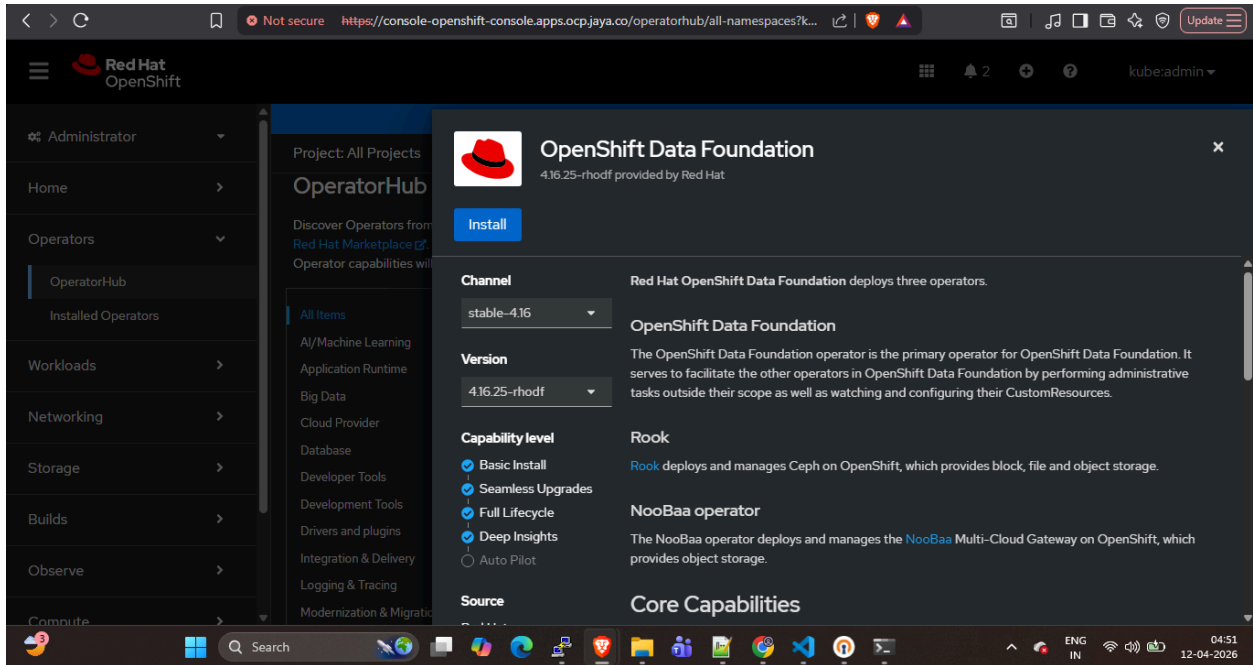
Step 2: Search Local storage and ODF

Search for:

Local storage

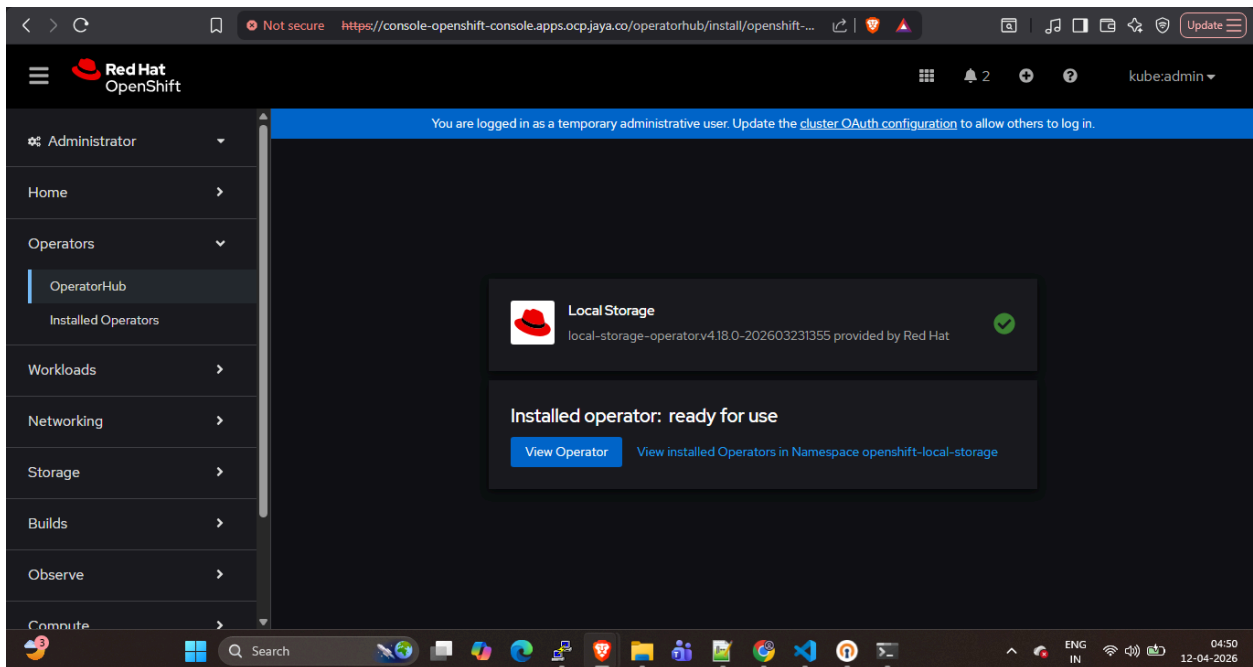
OpenShift Data Foundation

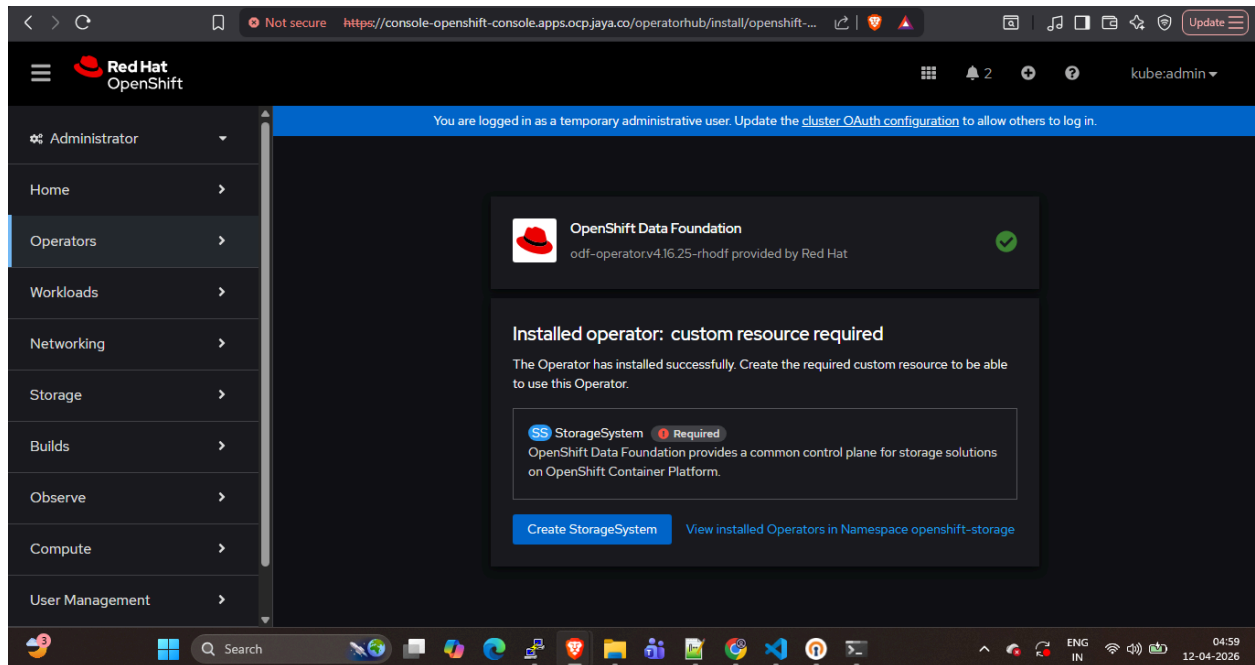




Step 3: Install Operator

- Update channel: stable
- Installation mode: All namespaces
- Approval: Automatic





I verified by following the below methods

```
[root@bastion install]# oc get pods -n openshift-storage
```

NAME	READY	STATUS	RESTARTS	AGE
csi-addons-controller-manager-5b87cfd955-x8s4f	2/2	Running	1 (2m47s ago)	3m50s
noobaa-operator-85cf956b5b-647zj	1/1	Running	0	4m45s
ocs-operator-848854984d-jvdtm	1/1	Running	1 (2m48s ago)	4m28s
odf-console-574cdf8795-kjb7t	1/1	Running	0	4m22s
odf-operator-controller-manager-6b7bb564-6jq9h	2/2	Running	1 (2m47s ago)	4m22s
rook-ceph-operator-954bcb77c-qt4wh	1/1	Running	0	4m3s
ux-backend-server-5fcfb65bb7-d7jvm	2/2	Running	0	4m28s

```
[root@bastion install]#
```

```
[root@bastion install]# oc get deployment -n openshift-storage
```

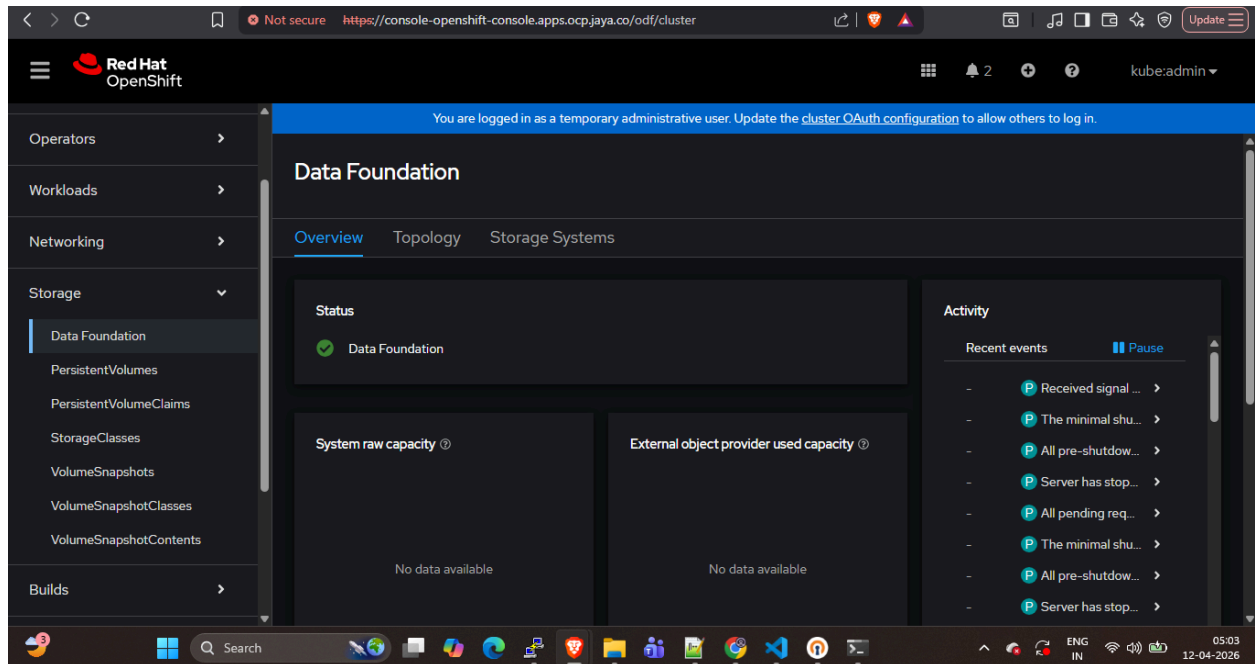
NAME	READY	UP-TO-DATE	AVAILABLE	AGE
csi-addons-controller-manager	1/1	1	1	5m9s
noobaa-operator	1/1	1	1	5m11s
ocs-client-operator-console	0/0	0	0	4m57s
ocs-client-operator-controller-manager	0/0	0	0	4m57s
ocs-operator	1/1	1	1	4m54s
odf-console	1/1	1	1	4m48s
odf-operator-controller-manager	1/1	1	1	4m48s
prometheus-operator	0/0	0	0	4m51s
rook-ceph-operator	1/1	1	1	4m30s
ux-backend-server	1/1	1	1	4m54s

```
[root@bastion install]#
```

6. Creating Storage System

Step 1: Navigate to Storage

OpenShift Console → Storage → Data Foundation



Step 2: Create StorageSystem

Click:

Create StorageSystem

Select:

- Full deployment

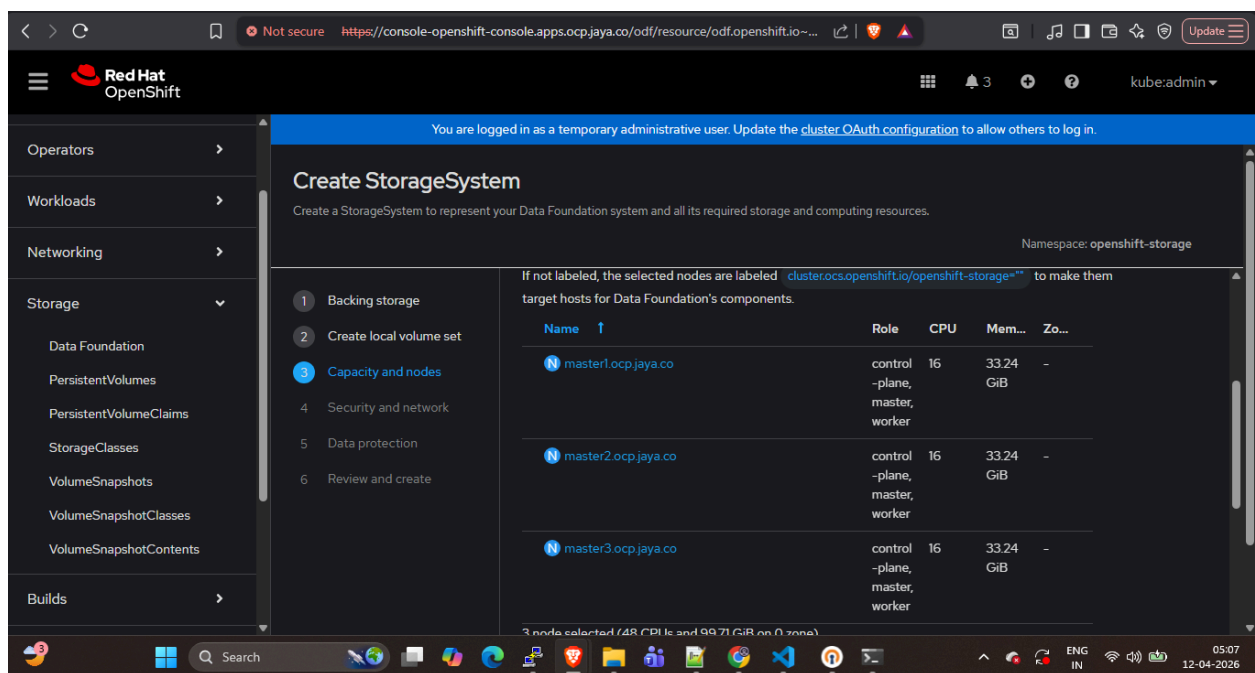
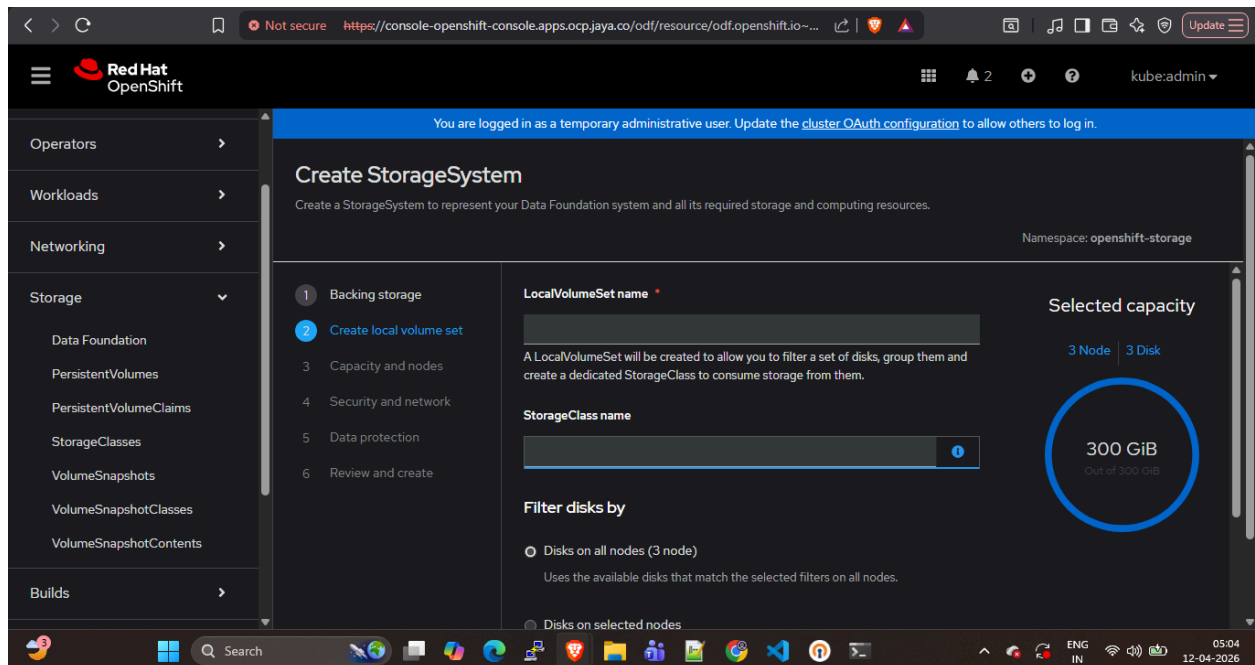
7. Node and Disk Selection

Step 1: Select Nodes

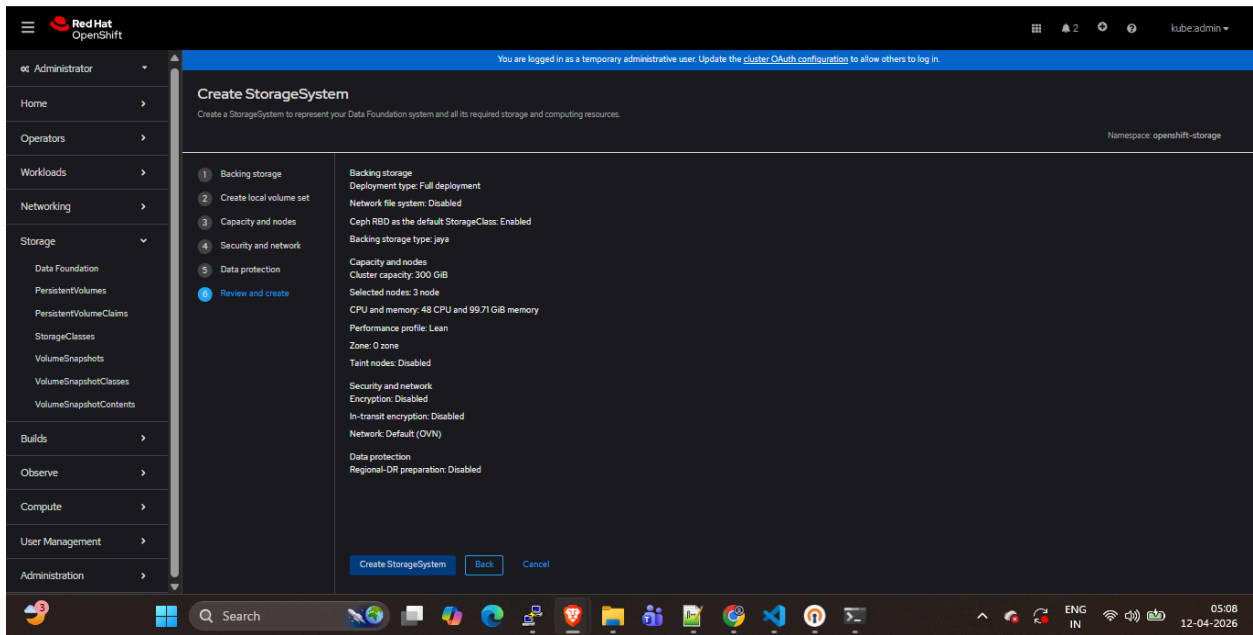
Click on create storage system

Step 2: Give name for LV

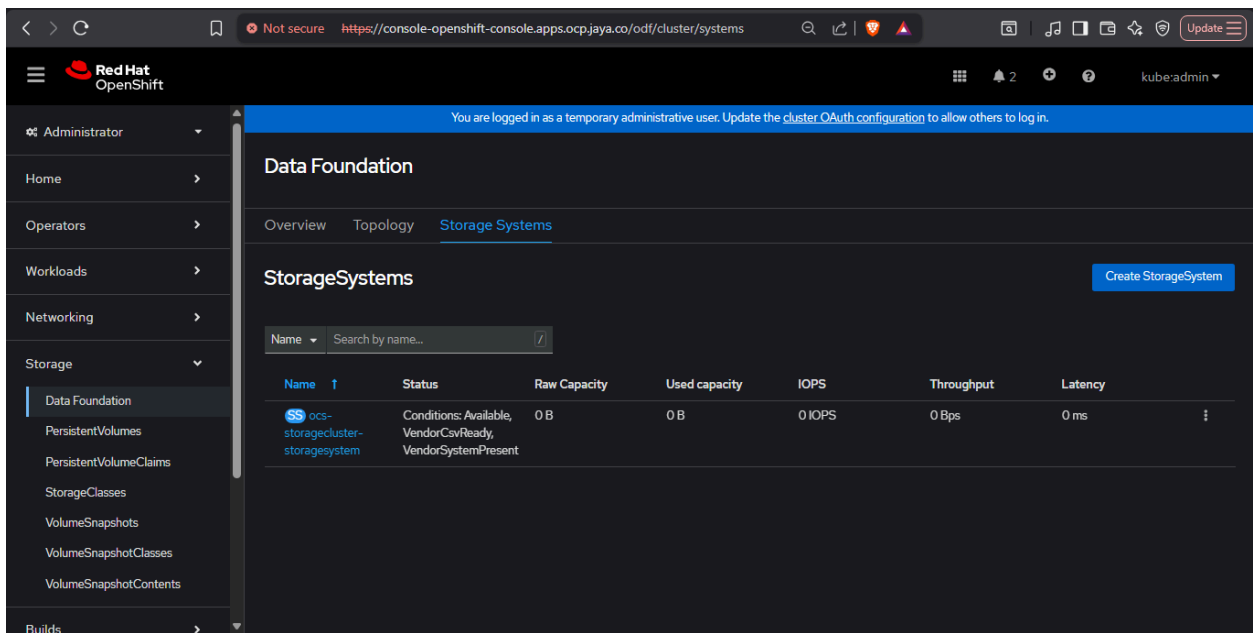
Step 3: ODF will detect new disks



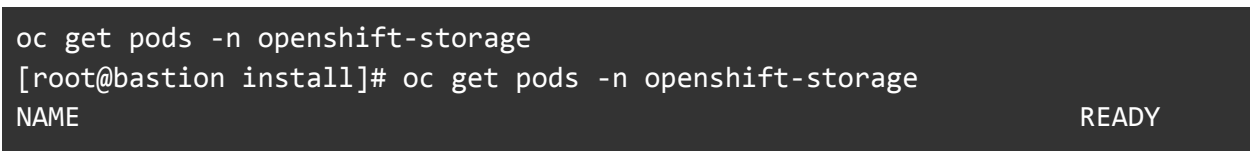
Step 4: follow through the process and review and final



8. Validation Steps



8.1 Check Pods



STATUS	RESTARTS	AGE	
			csi-addons-controller-manager-5b87cfd955-x8s4f 2/2
Running	2 (4m17s ago)	20m	
			csi-cephfsplugin-4bgtv 2/2
Running	0	5m27s	
			csi-cephfsplugin-97qdm 2/2
Running	0	5m27s	
			csi-cephfsplugin-c2994 2/2
Running	1 (4m42s ago)	5m27s	
			csi-cephfsplugin-provisioner-6b88d6db64-4gr2t 6/6
Running	0	5m27s	
			csi-cephfsplugin-provisioner-6b88d6db64-vl6cb 6/6
Running	2 (80s ago)	5m27s	
			csi-rbdplugin-4pb8k 3/3
Running	2 (66s ago)	5m27s	
			csi-rbdplugin-4wgsd 3/3
Running	0	5m27s	
			csi-rbdplugin-5xmdk 3/3
Running	1 (96s ago)	5m27s	
			csi-rbdplugin-provisioner-864ccb678c-4954z 6/6
Running	4 (63s ago)	5m27s	
			csi-rbdplugin-provisioner-864ccb678c-8rdbg 6/6
Running	2 (96s ago)	5m27s	
			noobaa-core-0 2/2
Running	0	45s	
			noobaa-db-pg-0 1/1
Running	0	45s	
			noobaa-operator-85cf956b5b-647zj 1/1
Running	0	21m	
			ocs-metrics-exporter-6566b87f65-h4w4j 1/1
Running	0	40s	
			ocs-operator-848854984d-jvdtm 1/1
Running	3 (4m3s ago)	20m	
			odf-console-574cdf8795-kjb7t 1/1
Running	0	20m	
			odf-operator-controller-manager-6b7bb564-6jq9h 2/2
Running	3 (4m24s ago)	20m	
			rook-ceph-crashcollector-master1.ocp.jaya.co-764d879556-cwdth 1/1
Running	0	2m6s	
			rook-ceph-crashcollector-master2.ocp.jaya.co-5cd89d7bf-88mwv 1/1
Running	0	68s	
			rook-ceph-crashcollector-master3.ocp.jaya.co-7747d57b-pxhms 1/1
Running	0	67s	

```

rook-ceph-exporter-master1.ocp.jaya.co-f7d5f7949-xqrw2      1/1
Running      0      2m3s
rook-ceph-exporter-master2.ocp.jaya.co-54f9486fbd-fg9n2      1/1
Running      0      65s
rook-ceph-exporter-master3.ocp.jaya.co-75bc867cf7-l5nzf      1/1
Running      0      63s
rook-ceph-mds-ocs-storagecluster-cephfilesystem-a-f5d8cbdfgt5p1 2/2
Running      0      68s
rook-ceph-mds-ocs-storagecluster-cephfilesystem-b-548b95592s68s 2/2
Running      0      67s
rook-ceph-mgr-a-56ff4c8d9d-4phs7      3/3
Running      0      2m43s
rook-ceph-mgr-b-67c696f7dc-dl2td      3/3
Running      0      2m42s
rook-ceph-mon-a-865969b7b5-jf648      2/2
Running      0      3m33s
rook-ceph-mon-b-c767796cc-6fcq2      2/2
Running      0      3m5s
rook-ceph-mon-c-6565d58984-dhxjs      2/2
Running      0      2m54s
rook-ceph-operator-954bcb77c-qt4wh      1/1
Running      0      20m
rook-ceph-osd-0-7c7ffb994c-k6r6h      2/2
Running      0      2m8s
rook-ceph-osd-1-ff57bd48f-2swsn      2/2
Running      0      2m7s
rook-ceph-osd-2-84945478b9-xmgsf      2/2
Running      0      2m7s
rook-ceph-osd-prepare-232cbabe63fb3107ab26caa9c37fea62-z4djl 0/1
Completed    0      2m20s
rook-ceph-osd-prepare-5153d2ee472cb0a2276ab76b75d839d7-ckz9c 0/1
Completed    0      2m20s
rook-ceph-osd-prepare-626003c651246775544c94f8a04837ce-qdfct 0/1
Completed    0      2m20s
rook-ceph-rgw-ocs-storagecluster-cephobjectstore-a-86ddc4frsnhj 2/2
Running      0      47s
ux-backend-server-5fcfb65bb7-d7jvm      2/2
Running      0      20m
[root@bastion install]#

```

8.2 Check StorageCluster

```
oc get storagecluster -n openshift-storage
[root@bastion install]# oc get storagecluster -n openshift-storage
NAME                                AGE    PHASE    EXTERNAL    CREATED AT
VERSION
ocs-storagecluster                 6m     Progressing
4.16.25                             2026-04-11T23:38:43Z
```

Expected:

- Phase: Ready

8.3 Check StorageClasses

```
oc get sc
[root@bastion install]# oc get sc
NAME                                PROVISIONER
RECLAIMPOLICY    VOLUMEBINDINGMODE    ALLOWVOLUMEEXPANSION    AGE
jaya
Delete          WaitForFirstConsumer  false                    9m14s
ocs-storagecluster-ceph-rbd (default)  openshift-storage.rbd.csi.ceph.com
Delete          Immediate             true                     2m5s
ocs-storagecluster-ceph-rgw
openshift-storage.ceph.rook.io/bucket  Delete                Immediate
false                    6m32s
ocs-storagecluster-cephfs
openshift-storage.cephfs.csi.ceph.com  Delete                Immediate
true                    109s
openshift-storage.noobaa.io
Delete          Immediate             false                    30s
[root@bastion install]#
```

